

BROTEK Multi-function RF admittance switches use a high-frequency radio signal to measure the changes in impedance caused by the material in contact with the probe. The RF admittance switch can detect the level of various types of materials with different dielectric constants, including solids, liquids, and powders. They are also resistant to the build-up of material on the sensing probe and can be used in hazardous environments. It is a new generation RF Admittance Level Transmitter with full independent property rights developed by Brotek after many years of hard work.

BT-RFS-XXXX



Technical Details

Power supply : 12 ~ 30VDC / 110 ~ 220VAC
Output : Relay DPDT 8A 250VAC
Probe length : User Specified
Connection method : Thread/Flange (user specified)
Environmental Humidity : $\leq 95\%$ RH
Relative to dielectric constant : $\epsilon \geq 1.5$
Measurement range : 0~2500 pF
Resolution : 1.0 pF
Response Time : 0.25 seconds
Working Temperature : Standard Type: -20~ 80°C,
High Temperature Type: -25 ~ 180 °C
Working Pressure : 0.1 ~ 10MPa
Switching delay : adjustable from 0 to 30 seconds
Switch mode : upper or lower limit alarm (user set)
Enclosure Material : die-cast aluminum alloy
(IP 66) (EX d II CT6)
Fork body material: 316 stainless steel / Teflon coated

Features

- Real-time display of capacitance value.
- User can choose low and high level alarm in single switch.
- High reliability and sensitivity(0.1pF)
- Wide range of materials(conductive and non conductive) can detect.
- It can overcome the influence of probe adhesion and hanging material on measurement.
- Easy to set: Sensitivity and alarm delay.
- High response time



Applications

RF admittance level switch can be used with a wide range of materials such as liquid powder, granular material and slurries. It can be used with conductive and non conductive materials, and are not affected by change in temperature, pressure, or viscosity. Some examples are Chemical and petrochemical processing plants and refineries, Food and beverage processing, Water treatment plants, Pharmaceutical manufacturing, Wastewater treatment plants, Mining and material handling.

